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BUILDING A SUSTAINABLE FUTURE TOGETHER

The G20 must set up a group of government and industry representatives to formulate green policies for steel

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Decarbonising steel will be pivotal to liberating India from its heavy and historic reliance on fossil fuels

INDIA'S G20 PRESIDENCY holds immense significance in driving meaningful progress on climate change. Addressing the G20 Environment and Climate Ministers' meeting in Chennai, Prime Minister Narendra Modi had recently asserted that climate action must follow '*Antyodaya*', which means that climate policies and initiatives must be designed in a way that they not only address environmental concerns but also social inequalities, and promote inclusivity.

The G20 must constitute a high-level group of government and industry representatives from each member for global co-operation, which will help in formulating green, public and corporate procurement policies to accelerate the development of net-zero steel. Establishing an actionable steel decarbonisation road map with considerations for national circumstances, technological innovation and capacity-building is essential. While steel production has historically been linked to emissions, there is a sense of optimism for the future. If steel changes, India changes. With a ripple of allied green manufacturing practices, the path to net-zero is becoming clearer. Further, achieving the \$5-trillion economy target by 2027 and doubling steel production capacity to 300 million tonnes in the next nine years are also intertwined. Which brings us to the question: Will India seize the opportunity to decarbonise steel manufacturing through '*Amrit Kaal*' and become '*Aatmanirbhar*'?

Accelerating Adoption of Renewable Energy

Decarbonising steel will be pivotal to liberating India from its reliance on fossil fuels. The government's target of 500 GW of installed renewable energy (RE) capacity by 2030 represents a concrete policy push towards this ambition. Achieving this necessitates a comprehensive approach that addresses both environmental and economic challenges concurrently, resulting in the accelerated adoption of RE sources like solar, wind, hydro, biomass and nuclear power. These sources can replace India's fossil fuel-fired power plants, which emit 65 per cent of CO₂.

Further, the development of power storage solutions for RE is also of utmost importance. At present, the installation of battery energy storage systems (BESS) is not economically viable. To bridge this gap, promoting the use of pumped hydro storage becomes crucial. Currently, the cost of green hydrogen remains high due to its primary expense: clean energy, which is also steeply priced, contributing 70 per cent to the total production cost. Hence, it becomes important to reduce the overall cost of RE. Also, electrolytes make up 15 per cent of the hydrogen energy cost,